

CSS

Position , pseudo class & elements

Theory

1. CSS Positioning

- **Static** → default, elements flow normally.
- **Relative** → positioned relative to its normal location (top, left, etc.).
- **Absolute** → positioned relative to the nearest positioned ancestor.
- **Fixed** → positioned relative to the viewport, stays fixed during scroll.
- **Sticky** → toggles between relative and fixed depending on scroll position.

2. Pseudo-classes

- **Definition:** Define special states of elements.
- Examples:
 - `:hover` → when mouse hovers.
 - `:focus` → when input is active.
 - `:first-child`, `:last-child`, `:nth-child(n)` → target specific list items.
 - `:link`, `:visited`, `:active` → link states.

3. Pseudo-elements

- **Definition:** Style specific parts of an element.
- Examples:
 - `::first-letter` → style first letter.
 - `::first-line` → style first line.
 - `::before` / `::after` → insert content before/after element.
 - `::selection` → style highlighted text.
 - `::marker` → style list markers.

4. Practical Use Cases

- Positioning → sticky navbars, fixed headers, tooltips.
- Pseudo-classes → interactive buttons, styled links.

- Pseudo-elements → decorative icons, custom list markers, styled text highlights.

Coding Round Questions

- Positioning → Explain difference between relative, absolute, fixed, and sticky positioning.
- Hover effect → Create a button that changes color on hover.
- Focus state → Style input fields when focused.
- Pseudo-elements → Use `::before` and `::after` to add icons to list items.
- Nth-child → Style even and odd rows differently in a table.
- Sticky nav → Build a sticky navigation bar using `position: sticky`.

⚡ Machine Coding Round Challenge

Problem: Interactive Learning Page

Requirements:

- Create a **sticky navigation bar** at the top.
- Add a **fixed header** that stays visible during scroll.
- Use **absolute positioning** for a floating box.
- Add a list of topics styled with pseudo-classes (:first-child, :nth-child).
- Use pseudo-elements (::before, ::after, ::marker) to decorate list items.
- Add hover and focus effects for interactivity.

Position.html

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Position</title>
    <style>
      div {
        width: 150px;
        height: 150px;
        border: 2px solid black;
        margin-bottom: 20px;
      }
      .relative-box {
        /* position: relative; */

```

```

    /* top , bottom, left ,right */
    /* top: 100px; */
    /* left: 100px; */
    background-color: blueviolet;
}

.absolute-box {
    position: absolute;
    background-color: gray;
    color: white;
    left: 200px;
    top: 20px;
}
.fixed-header {
    position: fixed;
    /* left: 200px; */

    background-color: yellow;
}
.sticky-nav{
    position: sticky;
    top: 0;
    z-index: -10;
    background-color: blanchedalmond;
}
</style>
</head>
<body>
    <div class="relative-box">Relative box</div>

    <div class="fixed-header">Fixed header</div>
    <div class="absolute-box">Absolute box</div>
    <div class="sticky-nav">sticky nav</div>

    <p>
        s labore tempora a incidunt dolorum. Dicta consequatur
        minus odit sunt rerum quaerat autem ipsa voluptatem perferendis id.
        Quisquam necessitatibus velit facilis eaque aut amet dignissimos sint
        assumenda porro cumque quod magnam, consequatur ipsa provident qui
        debitis
        reprehenderit modi repudiandae voluptates animi. Eum sapiente unde illum
        aut inventore?
    </p>
</body>
</html>

```

```
<!doctype html>
```

```
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Pseudo</title>
    <style>
      h1 {
        color: greenyellow;
      }

      h1:hover {
        color: blue;
        /* font-size: 60px; */
      }
      a:link {
        color: red;
      }
      a:visited {
        color: aquamarine;
      }
      a:active {
        color: yellow;
      }

      input:focus {
        outline: 2px solid orange;
      }

      ul li:first-child {
        color: violet;
        font-size: 20px;
      }

      ul li:last-child {
        color: tomato;
      }
      /* ul li:nth-child(6){
        color:aqua;
      } */

      ul li:nth-child(even) {
        color: aqua;
      }

      /* pseudo elements */
      /* p::first-letter{
        font-size: 60px;
        color: red;
      }
    </style>
  </head>
  <body>
    <h1>Pseudo</h1>
    <a href="#">Link</a>
    <a href="#">Visited</a>
    <a href="#">Active</a>
    <input type="text" />
    <ul>
      <li>First</li>
      <li>Second</li>
      <li>Third</li>
      <li>Fourth</li>
      <li>Fifth</li>
      <li>Sixth</li>
      <li>Seventh</li>
      <li>Eighth</li>
      <li>Ninth</li>
      <li>Tenth</li>
    </ul>
    <p>Pseudo</p>
  </body>
</html>
```

```

} */

p::first-line{
  color: red;
}

ul li::before{
  content: "😄";
  /* content: "inrctc "; */
}
::selection{
  background-color: chartreuse;
  color: blue;
}

li::marker{
  content: "✓";
}

ul li::after{
  content: "😄";
}
input::placeholder{
  color: rebeccapurple;
}
</style>
</head>
<body>
<p>
  Lorem ipsum dolor sit amet consectetur adipisicing elit. Libero, aliquam
  delectus optio autem, voluptate blanditiis, eos asperiores quo magni
  maxime eius molestias obcaecati aspernatur! Dignissimos culpa reiciendis
  vero laborum omnis?
</p>

<ul>
  <li>item 1</li>
  <li>item 2</li>
  <li>item 3</li>
  <li>item 4</li>
  <li>item 5</li>
  <li>item 6</li>
  <li>item 7</li>
  <li>item 8</li>
  <li>item 9</li>
  <li>item 10</li>
</ul>

```

```
<input type="text" placeholder="Enter your name"/>
<a href="#">Google</a>
<h1>Pseudo</h1>
</body>
</html>
```